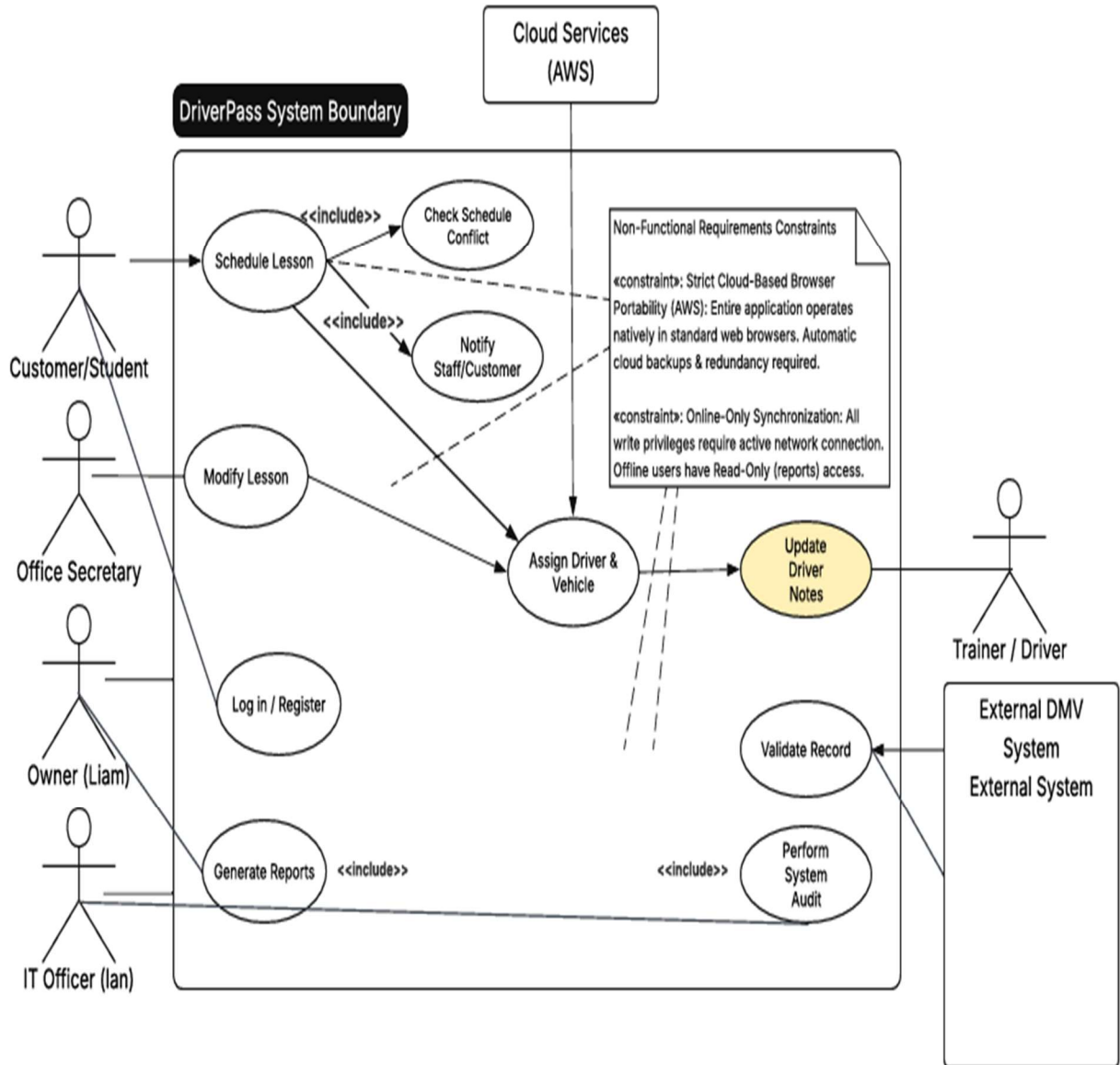


CS 255 System Design Document Template

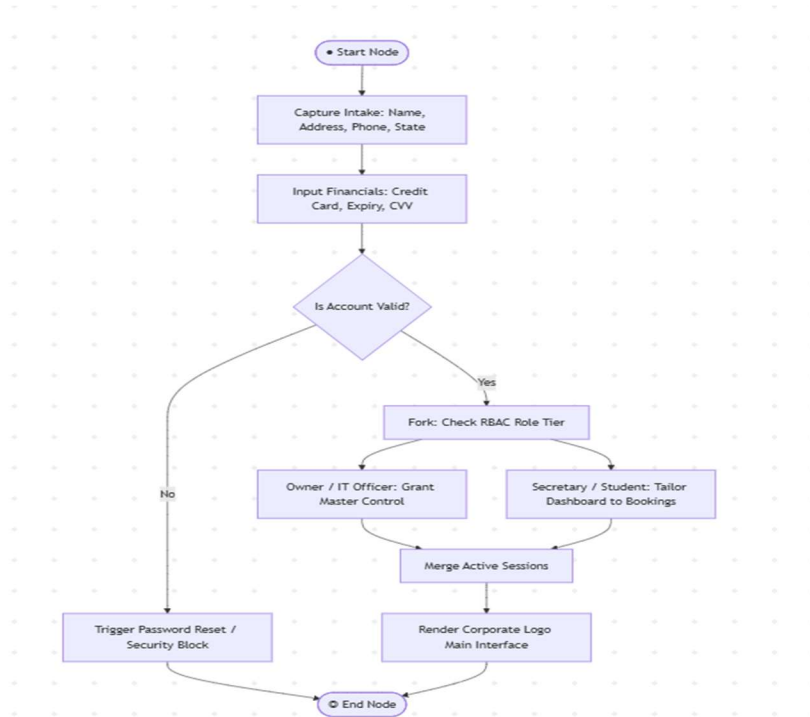
UML Diagrams

UML Use Case Diagram

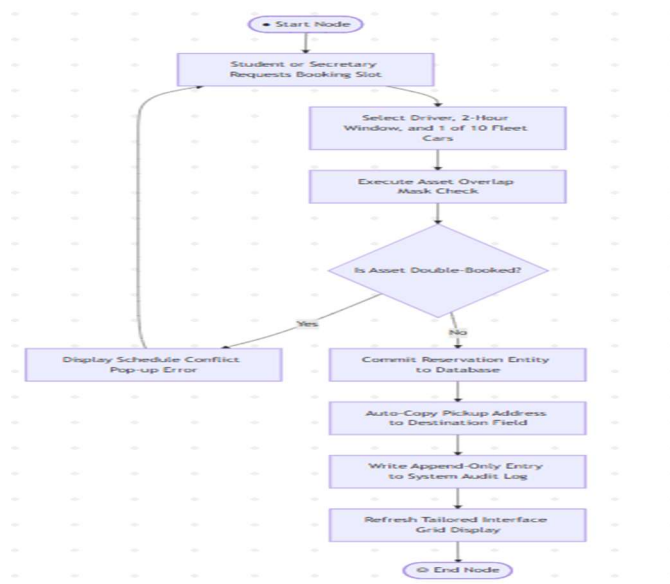


UML Activity Diagrams

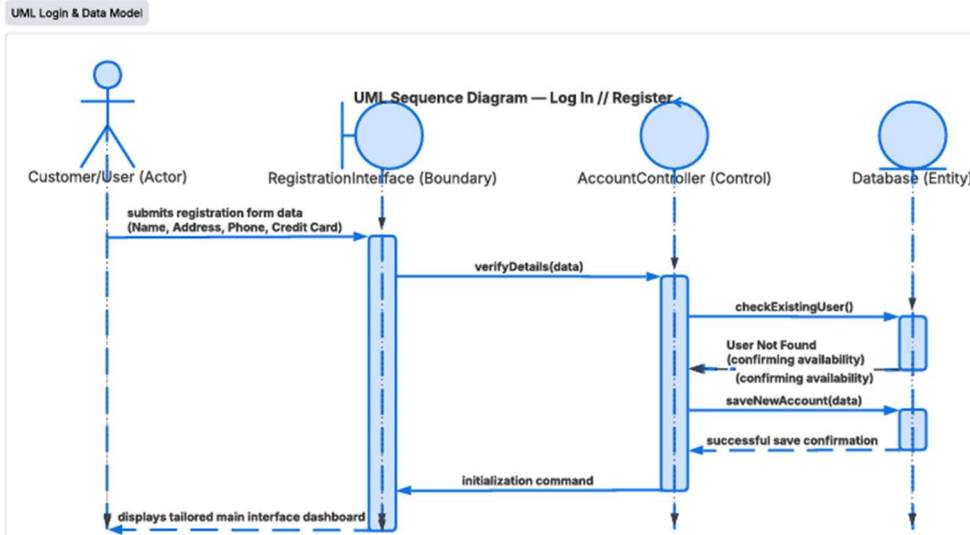
DriverPass Diagram 1: Use Case — Log In / Register



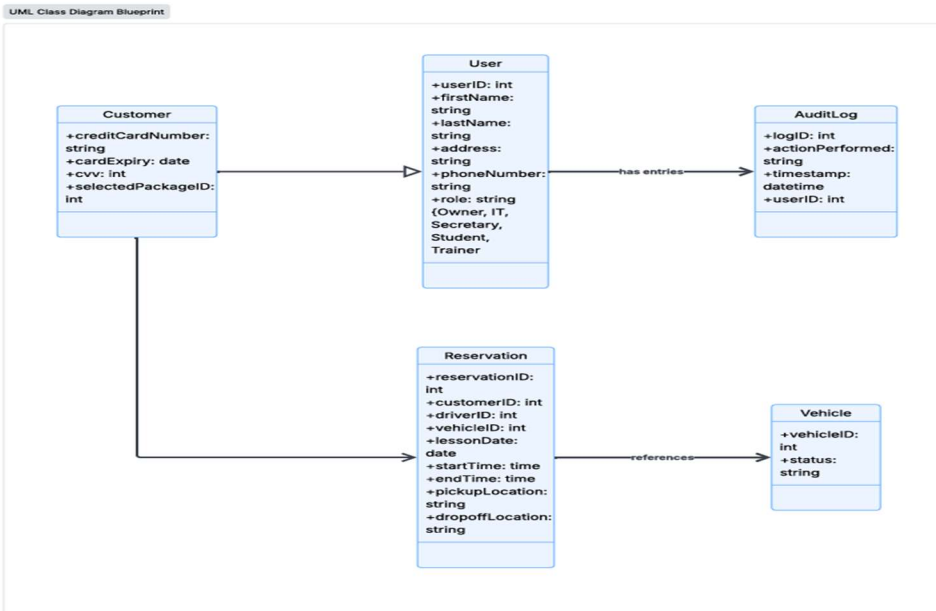
DriverPass Diagram 2: Use Case — Schedule Lesson



UML Sequence Diagram



UML Class Diagram



Technical Requirements

1. Hardware Environment and Infrastructure Foundation

- **Managed Cloud Implementation:** To completely bypass the capital expense and logistical difficulties of procuring, housing, and maintaining physical server hardware on-site, the core platform relies on a cloud-hosted infrastructure (e.g., Amazon Web Services).
- **High-Availability Failovers:** The application deployment architecture distributes operational tasks across geographically isolated availability zones, creating automatic recovery triggers that preserve runtime functionality during local server farm outages.

- **Client Terminal Specifications:** Fleet managers, administrative office employees, certified trainers, and student drivers can interface with the dashboard using conventional consumer computing hardware, including baseline laptops, desktop machines, mobile phones, or tablets connected to the internet.

2. Software Ecosystem, Core Platforms, and Integrations

- **Browser-Agnostic Web Portal:** The consumer-facing application relies on responsive web architectures that function natively inside contemporary web browsers (such as Safari, Edge, Chrome, and Firefox), eliminating the need for local desktop downloads.
- **Relational Database Management (RDBMS):** Transaction and data storage mechanisms use a cloud-managed database service (like Amazon RDS utilizing PostgreSQL or MySQL) to guarantee relational integrity, track simultaneous user inquiries, and log administrative updates.
- **Automated External Syncing:** The framework establishes real-time connections to the Department of Motor Vehicles (DMV) registries via secure REST API portals, enabling automatic synchronization of updated state driving mandates, policy rules, and testing materials.

3. Dedicated Utilities and Core Operational Capabilities

- **Analytical Extraction Modules:** Native data-handling components allow administrative users to pack scheduling parameters, ledger audits, and instructor mileage metrics into spreadsheet formats like .xlsx or .csv files for deep business parsing.
- **Granular Identity Management:** System boundaries enforce strict User Privilege Mapping (UPM), which automatically restricts access parameters depending on account categories (such as Student, Trainer, Clerical Staff, or Business Executive) to prevent unauthorized information access.
- **Nightly Snapshot Routines:** Automated recovery mechanisms execute full-system database backups during low-activity midnight windows, protecting the enterprise from data loss by providing clean restoration options in case of runtime corruption events.